

PAPER160 — Ug4 Extended: Vacuum Concentration, AGN Feedback, and $\rho_v=6e-27$ Calibration

Session: 47 | Date: March 13, 2026 | Thread: 7f9068 | Domain: §2.3

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Abstract

This paper documents three new calibration parameters for the UQFF Ug4 term: vacuum energy density $\rho_v = 6 \times 10^{-27} \text{ kg/m}^3$, vacuum concentration factor $C_{\text{concentration}} = 1.0$, and AGN/stellar feedback factor $f_{\text{feedback}} = 0.1$. These extend PAPER086 (Ug4 AGN Feedback 8-parameter formula) with physical calibrations confirmed in Grok thread 7f9068 C++ execution.

1. Original Ug4 Formulation (PAPER_086 Reference)

From §1.11 PAPER_086:

$$U_{\text{g4}}(t, t_n) = k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot \frac{M_{\text{bh}}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{feedback}})$$

PAPER086 documented this as an 8-parameter formula but did not calibrate ρ_v , $C_{\text{concentration}}$, or f_{feedback} . These three parameters were undefined/defaulted in the §2.2 implementation.

2. New Calibrations (Thread 7f9068)

2.1 Vacuum Energy Density: $\rho_v = 6 \times 10^{-27} \text{ kg/m}^3$

Source: NIST CODATA 2022 + cosmological vacuum energy measurements (Planck 2018 Ω_Λ).

The observed dark energy density:

$$\rho_\Lambda = \frac{\Lambda c^2}{8\pi G} \approx 5.96 \times 10^{-27} \text{ kg/m}^3$$

Calibrated to $6 \times 10^{-27} \text{ kg/m}^3$ in the UQFF vacuum concentration term.

2.2 Vacuum Concentration Factor: $C_{\text{concentration}} = 1.0$

Physical interpretation: C_{conc} modulates how much vacuum energy is "concentrated" near the black hole/galactic center relative to the cosmic mean. $C_{\text{conc}} = 1.0$ = isotropic baseline (no enhancement). Expected range: 0.1–100 for AGN environments.

2.3 AGN/Stellar Feedback Factor: $f_{\text{feedback}} = 0.1$

Physical interpretation: AGN jets + stellar winds inject energy into the vacuum, increasing the effective Ug4 by ~10% ($f_{\text{feedback}} = 0.1 \rightarrow 1 + 0.1 = 1.1 \times \text{multiplier}$). Derived from observed AGN feedback efficiency $\epsilon_{\text{feedback}} \sim 0.05\text{--}0.15$ (mean 0.10).

3. Extended Ug4 Equation

$$\boxed{U_{\text{g4}}(t,t_n) = k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot \frac{M_{\text{bh}}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{\text{feedback}})}$$

With calibrated values at $t=0, t_n=0$:

$$U_{\text{g4}}(0,0) = 2.0 \times 6 \times 10^{-27} \times 1.0 \times \frac{8.15 \times 10^{36}}{2.55 \times 10^{20}} \times 1.0 \times 1.0 \times 1.1$$

$$= 2.0 \times 6 \times 10^{-27} \times 3.196 \times 10^{16} \times 1.1$$

$$\approx 4.219 \times 10^{-10} \text{, } \text{m/s}^2$$

This matches the computed $U_{\text{g4}} = 4.219 \times 10^{-10}$ for all four Solar System bodies (uniform at $t=0$ since it depends only on global M_{bh}/d_g , not per-body mass).

4. Parameter Inventory

Parameter	New Value	Prior Value	Physical Basis
ρ_v	$6 \times 10^{-27} \text{ kg/m}^3$	undefined	Planck 2018 Ω_Λ dark energy density
$C_{\text{concentration}}$	1.0	undefined	Isotropic vacuum baseline
f_{feedback}	0.1	undefined	AGN efficiency $\epsilon \sim 0.10$ (observed)
k_4	2.0	2.0 (unchanged)	UQFF canonical
M_{bh}	$8.15 \times 10^{36} \text{ kg}$	same	SgrA* black hole mass
d_g	$2.55 \times 10^{20} \text{ m}$	same	Distance to galactic center

5. Implications for UQFF Solvability

The calibration $\rho_v = \Lambda c^2/(8\pi G)$ establishes a **direct bridge** between UQFF Ug4 and the Λ CDM cosmological constant, completing the dark energy chain:

$$\Lambda \cdot c^2/3 \quad \xleftrightarrow{\text{PAPER_160}} \quad k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot M_{\text{bh}}/d_g$$

The compressed cosmological term $\Lambda c^2/3$ in PAPER_090 and the Ug4 vacuum term here are **complementary** representations of the same dark energy at different scales (global vs. local).

6. CP Integration

CP2 update: UQFFug4AGNFeedbackCalculator — add $C_{\text{concentration}}$, f_{feedback} , ρ_v parameters with defaults matching calibration.

CP3 update: FU_SolarSystem_*_Calculator — Ug4 component uses these calibrations.

Status: ■ Complete | **CP Stage:** CP2/CP3 **Supersedes:** N/A (extends PAPER086) | **Related:** PAPER086 (Ug4 AGN), PAPER090 (*compressed cosmological term*), PAPER106 (vacuum energy)

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \exp(-\tau) = 1 - 5.7e-1$ ■ $\exp(-2.9e-4) = 4.3e-1$; F_U at event horizon = $2.0e+18$ m/s■.